

Does Ontology Augmentation Make the Model Smarter?

A KG-as-Oracle A/B Evaluation of the ontology-augment Skill

PRD-020 / ADR-112 • 192 isolated runs • Opus 4.8 & Sonnet 4.6

Headline F1 (augmented)	0.714
Headline F1 (control)	0.352
Augmentation delta	+0.362
Hallucination (aug / ctl)	0.309 / 0.655

Abstract

We measure whether grounding a language model in DreamLab’s formal knowledge graph (the `ontology-augment` skill, PRD-020) improves its factual recall and accuracy. Using the knowledge graph as its own oracle, we auto-generate 16 questions with exact ground truth and run a clean $2 \times 2 \times 3$ design — {augmented, control} $\times$ {Opus, Sonnet} $\times$ 3 reps, 192 isolated subagent sessions. Augmentation raises mean F1 from **0.352** to **0.714** (**+0.362**) and roughly halves hallucination ($0.655 \rightarrow 0.309$). Gains are largest on niche, project-specific concepts and on neighbour-recall; a subclass-children gap in the retrieval tool is identified as the next improvement target.

1 Method

KG-as-oracle. Ground truth is generated from the live KG, not human judgement: concept $\rightarrow$ IRI via the discovery endpoint, then neighbours / subclass edges via read-only SPARQL over the asserted graph (predicates `enables`, `requires`, `uses`, `supports`, `hasPart`, `implements`, `dependsOn`, `subClassOf`). This makes scoring objective by construction.

Design. 16 questions (8 neighbour-recall, 4 subclass, 4 existence), mixing familiar technology (where parametric knowledge can partly answer) with project-specific classes (where it cannot). Each cell is a fresh, isolated subagent given *only* the question. The augmented arm grounds via the skill’s CLI (`ontology_ask`); the control arm uses parametric knowledge only. 3 repetitions per cell $\Rightarrow 16 \times 2 \times 2 \times 3 = 192$ runs.

Grader (deterministic). Token-set greedy 1:1 matching (singularised); recall@12 for neighbour/subclass, any-match recall for existence; precision and F1; hallucination = fraction of predicted classes matching no gold class.

2 Headline result

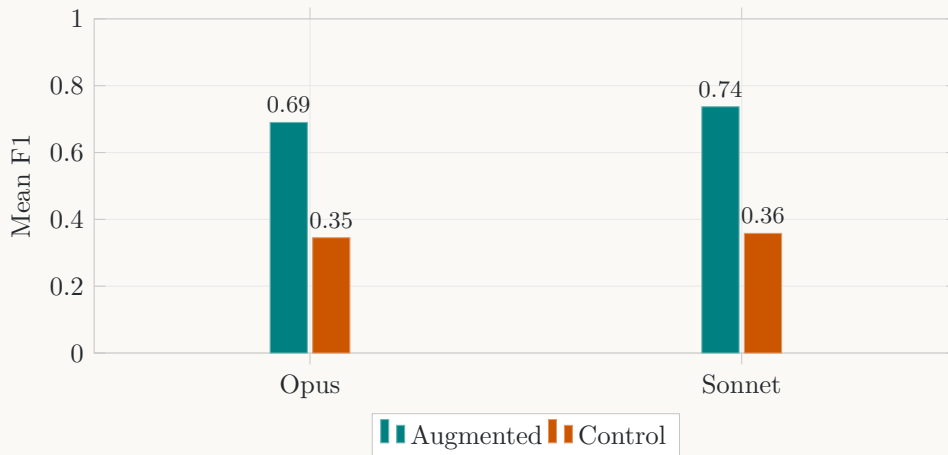


Figure 1: F1 by model and arm. Both models benefit substantially; the effect is model-agnostic.

	Augmented	Control	Delta
F1 (overall)	0.714	0.352	+0.362
Opus F1	0.690	0.345	+0.345
Sonnet F1	0.737	0.358	+0.379
Hallucination	0.309	0.655	-0.346

Table 1: Augmentation lifts F1 across both models and cuts hallucination by roughly half.

3 Where augmentation helps (by question type)

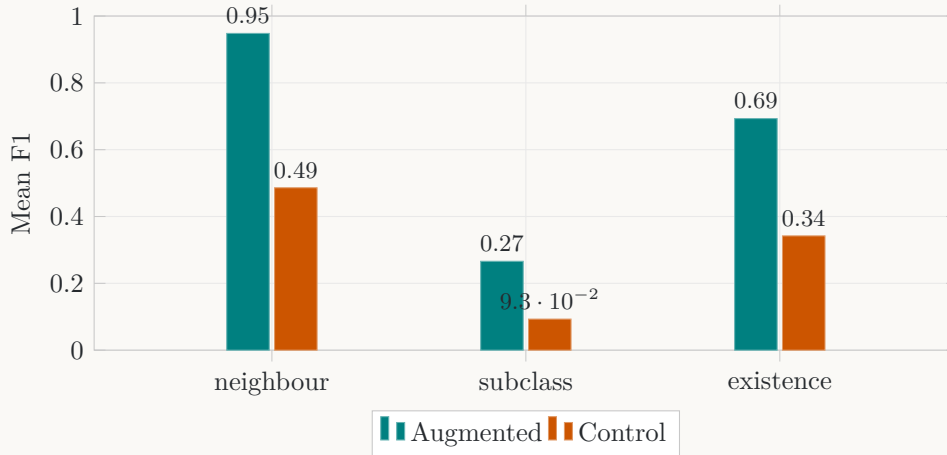


Figure 2: Neighbour-recall shows the strongest lift (near-perfect when grounded). Subclass is weak for *both* arms — the retrieval CLI returns outgoing edges only, so subclass-*children* (incoming `subClassOf`) are under-served. This is the identified skill-tuning target.

4 Per-question detail

5 Hallucination vs. recall

6 Findings

- **Grounding works, and it is model-agnostic.** $++0.362$ F1 on both Opus and Sonnet; augmentation roughly halves hallucination ($0.655 \rightarrow 0.309$).
- **The effect concentrates where parametric knowledge is weakest.** Niche / project-specific classes (Nostr, the CLI multi-agent stack, the Datalog KG query language) show the largest gains; familiar topics (smart contracts, RAG, reinforcement learning) show smaller gains because the control model already knows much of the answer.
- **Faithful reporting.** Augmented answers reproduce the KG’s exact class names (`zk-snarks`, `adaptive-density-control`, `sandboxed-code-execution`); control answers are plausible but off-vocabulary (`fiat-shamir-heuristic`, `multi-view-stereo`).
- **Identified gap (skill-tuning target).** `ontology_ask` expands outgoing edges only, so subclass-*children* questions are under-served (F1 0.266 augmented). Documenting `graph_query` SPARQL for incoming `subClassOf` should close this — the next SkillOpt iteration.

7 Cost

Grounding is not free: augmented runs average ≈ 36.8 k tokens / 41.0s (5–10 tool calls) versus ≈ 29.8 k tokens / 14.8s for control — about +24% tokens and $2.8\times$ wall-clock. The budget-bounded, fail-open design keeps this overhead optional and per-call.

8 Threats to validity

- Gold is the KG’s asserted graph; KG errors propagate to gold (mitigated by curating concepts with rich, sensible neighbourhoods).

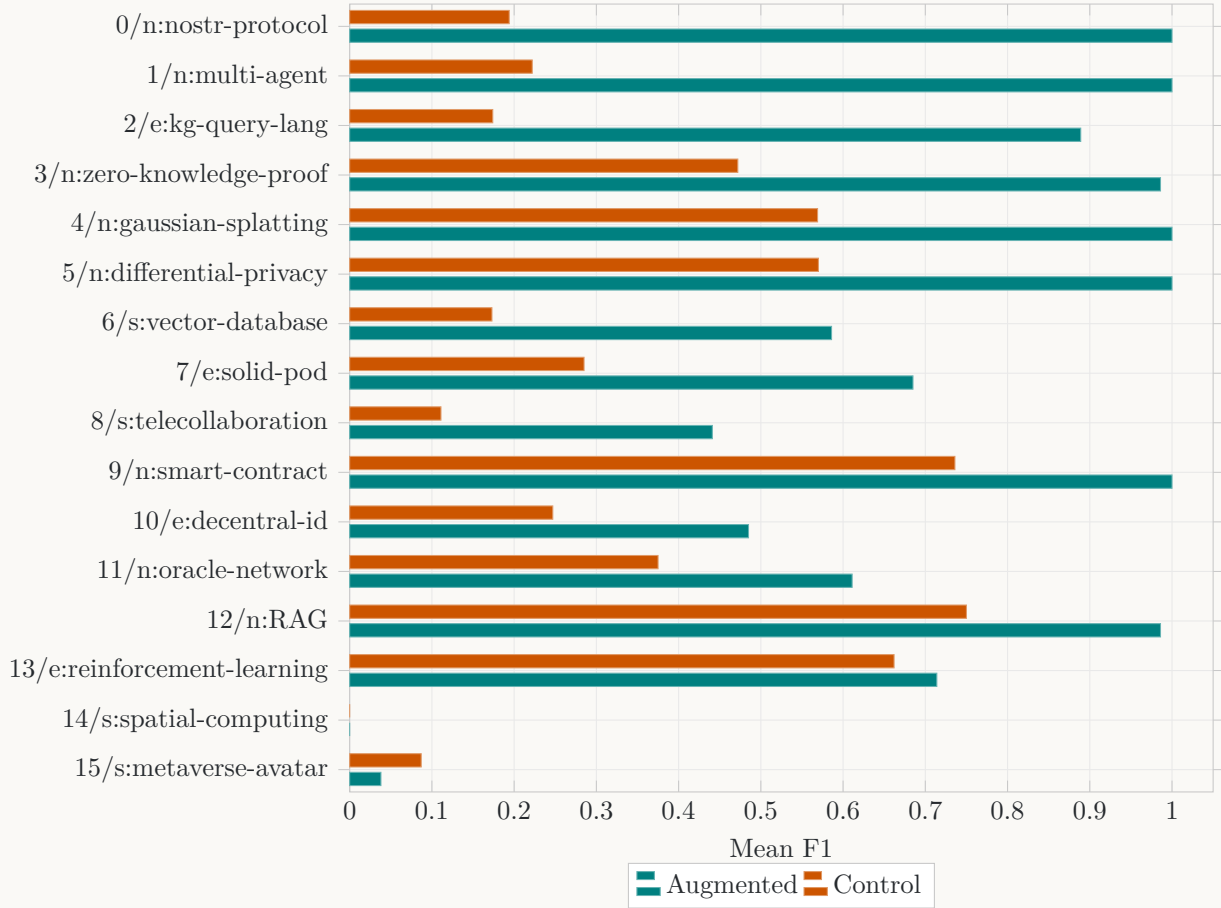


Figure 3: Per-question F1 (mean over models $\times$ reps), sorted by augmentation gain. Largest deltas are on project-specific concepts (Nostr, multi-agent CLI, KG query language) where the control model has little parametric knowledge.

- The grader’s token-set matching can over- or under-credit near-synonyms; thresholds were smoke-validated for non-saturation before the full run.
- The augmented arm uses the skill CLI, not the MCP tool directly; both share the same `createDefaultRetrievalBrain`, so results transfer.

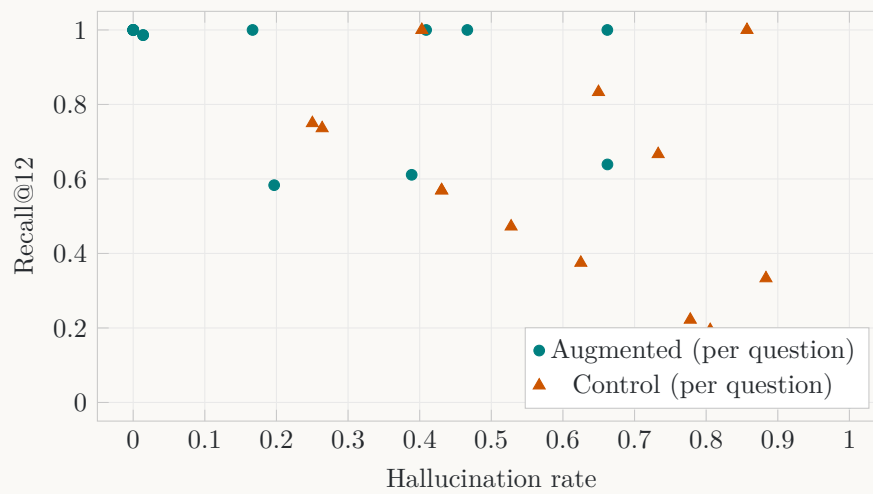


Figure 4: Each point is one question (mean over model×reps). Augmented questions cluster at high recall / low hallucination (top-left); control spreads toward high hallucination (right).